

Time Series Analysis and Forecasting

Honours 2026, lecture slides

Department of Statistical Sciences

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Foundations

Learning Outcomes

By the end of this chapter you should be able to:

- Explain what a time series is, and treat an observed series as one realisation of a stochastic process
- Judge predictability, and separate forecasts from goals and plans
- Identify trend, seasonality, cycle and irregular variation; produce the four benchmarks
- Use the lag/lead operators; define stationarity and white noise
- Identify MA, AR, ARMA, ARIMA, SARIMA from an ACF/PACF and forecast with Box–Jenkins
- Explain volatility clustering and fit ARCH/GARCH models

Notation and Conventions

Symbol	Meaning
y_t, n, T, h, m	Series, size, origin, horizon, cycle length
$\hat{y}_{T+h T}$	Forecast for $T + h$ given $\mathcal{I}_T$
e_{T+h}	$y_{T+h} - \hat{y}_{T+h T}$
ε_t	Innovation
L, F, ∇	Lag, lead, difference operators
$\gamma(h), \rho(h), \phi_{hh}$	Autocov., autocorr., partial ACF
ϕ, θ, σ_t^2	AR/MA coefficients; conditional variance

- x_t is always explanatory, never the series
- L = lag; some texts write B
- T =origin, T_t =trend, $\mathcal{T}$ =index set - keep apart
- Forecasts run from the series end, so $T = n$
- `ts(freq = m)` takes the **period** m , not $f = 1/m$

Why Ordering Matters

Modern institutions generate huge sequences of observations through time: exchange rates, electricity demand, GDP, heart rate, share prices, across economics, finance, medicine, epidemiology, environment, engineering, and ecology.

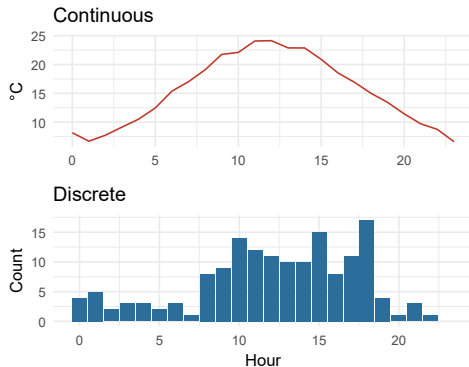
An ordered collection of observations recorded sequentially through time,

$$\{y_t : t = 1, 2, \dots, n\}.$$

- Ordinary regression assumes independent **errors**, not necessarily independent observations
- Time series analysis models the temporal dependence explicitly, rather than assuming it away - the subject matter changes across fields, the statistical problem does not

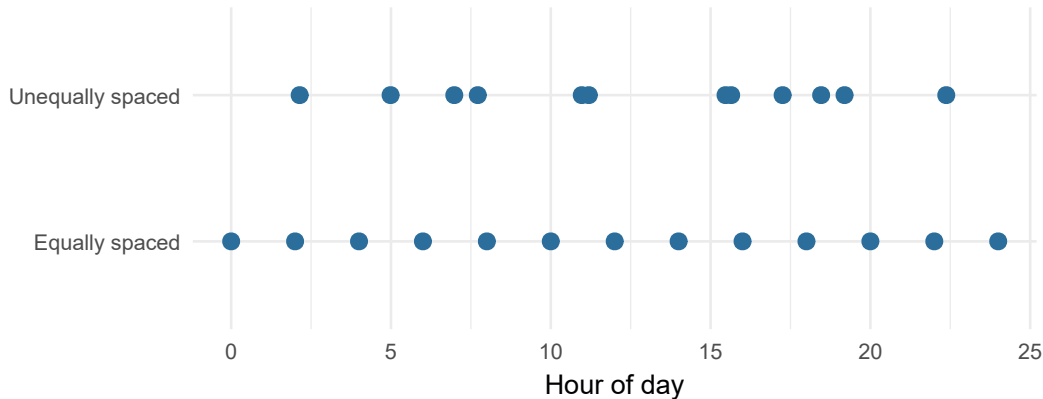
Qualitative, Continuous, and Discrete Series

- **Qualitative:** weather category, credit rating - no arithmetic, so no ACF; use a Markov chain instead
- **Continuous:** temperature - any real value, plotted as a line
- **Discrete:** arrival counts - non-negative integers, plotted as bars, modelled with Poisson/count methods



Equally vs Unequally Spaced Observations

A transaction log or hospital admission record fires whenever the event happens, not on a fixed clock. $t_s()$, ACF, and ARIMA all assume the regular case; irregular data need point-process models instead.



Cross-Sectional vs Time Series Data

- **Cross-sectional:** many units, one moment (e.g. grades of 40 students). Order carries no information.
- **Time series:** one process, many moments (e.g. annual GDP). Reordering destroys trend and seasonality.

	Cross-Sectional	Time Series
Unit	Many, one time	One, many times
Row order matters?	No	Yes, fundamentally
Typical assumption	Independence	Autocorrelation
"Neighbour"	Arbitrary	Yesterday, last week

Order Matters: A Demonstration

Same 24 numbers, shuffled. Reshuffling a cross-sectional dataset changes nothing; reshuffling a time series destroys the one thing that made it a time series - the trend disappears entirely.

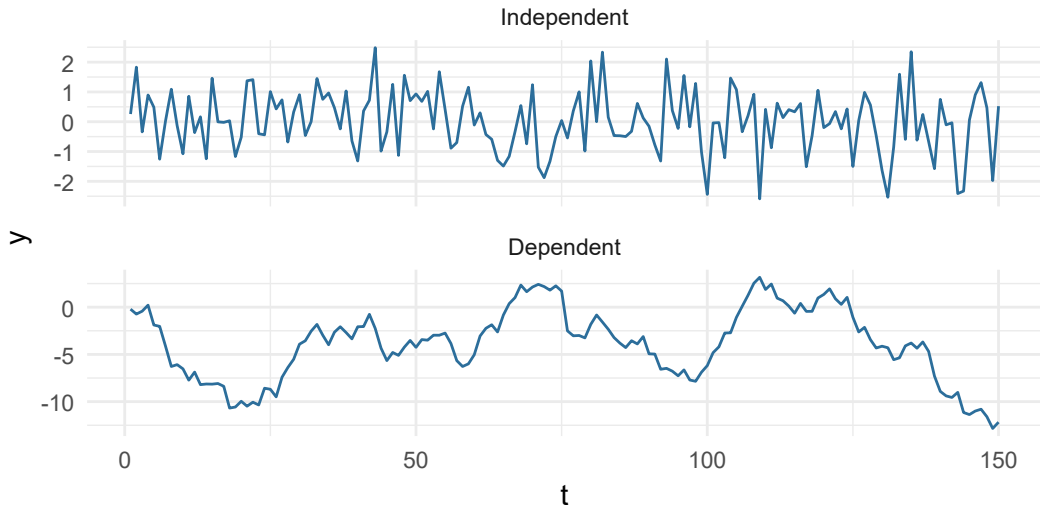
Original order



Shuffled



Dependence Made Visible



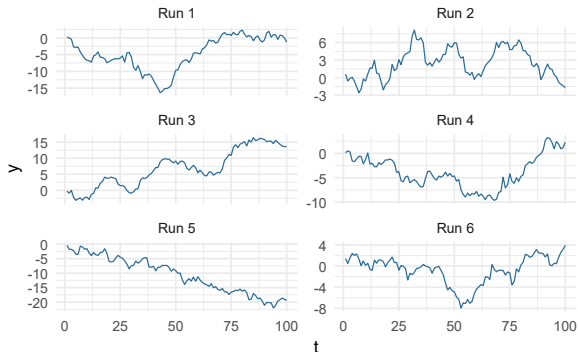
Temporal Granularity and the Aggregation Effect

Nothing about the underlying process changes across resolutions - only what you can see changes.

- Aggregating up **smooths out** high-frequency structure (a weekday dip is averaged away weekly)
- Aggregating up **reveals** low-frequency structure (the trend becomes clearer monthly)
- Whichever granularity you choose has already decided which patterns you are even capable of finding

One Realisation of a Stochastic Process

A collection of random variables indexed by time, $\{y_t : t \in \mathcal{T}\}$. Before observation each y_t is random; after observation it is realised. Six paths from one model - no two alike. Inference targets the **process**, not one dataset; a model reproducing one realisation exactly is almost certainly overfitted.



Deterministic and Stochastic Behaviour

- **Deterministic:** $y_t = 5 + 0.2t$ - fully known once t is known
- **Stochastic:** $y_t = 5 + 0.2t + \varepsilon_t$, $\varepsilon_t \sim N(0, 1)$ - never pinned down exactly

This motivates writing a series as **signal plus noise**,

$$y_t = s_t + \varepsilon_t.$$

Monthly electricity demand: $y_t = 250 + 12 \sin(2\pi t/12) + \varepsilon_t$. Constant 250 = average demand; sine term = seasonal signal (peak every 12 months, amplitude 12); ε_t absorbs weather and everything else.

The Forecasting Problem

What Forecasting Is

People have always wanted to know what comes next prophecy and astrology have been replaced by systematic analysis of data, probability, and models, but the underlying problem is unchanged.

The use of available information to estimate a quantity that has not yet been observed.

The goal is not certainty. It is the best possible estimate from available evidence, together with a realistic assessment of the surrounding uncertainty.

What Can Actually Be Forecast

Tomorrow's sunrise: nearly perfect. Tomorrow's lottery numbers: impossible. Most problems sit in between, decided by four questions:

1. Do we understand the factors that drive the process?
2. Do we have enough **relevant** historical data?
3. Will the important relationships hold in the future?
4. Will publishing the forecast change the outcome?

Always state the sampling interval **and** the horizon before calling a series predictable - daily rainfall tomorrow is forecastable; daily rainfall three weeks out is not, even though monthly totals stay seasonal.

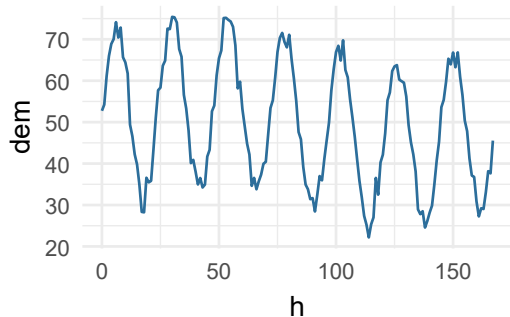
The Four Conditions

- **Understanding:** electricity demand responds to measurable drivers (temperature, calendar); asset prices respond to many unmeasurable ones
- **Relevant data:** more data is not automatically better data if the process has changed
- **Stability:** forecasting assumes main relationships stay reasonably stable, not that the future repeats exactly
- **Independence from the forecast:** a widely reported forecast can change the behaviour it predicts

Two Series, Same Length

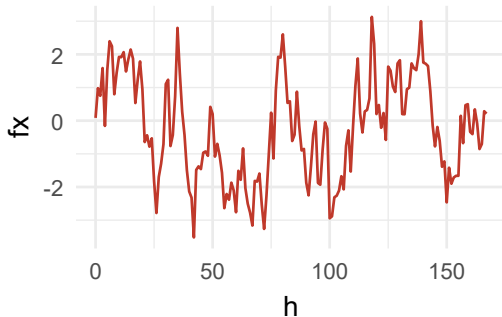
Electricity demand

Regular cycles



Exchange rate

Random walk



What matters is whether the data contain stable, informative patterns - not how many rows the spreadsheet holds.

Forecasting, Goals, and Planning

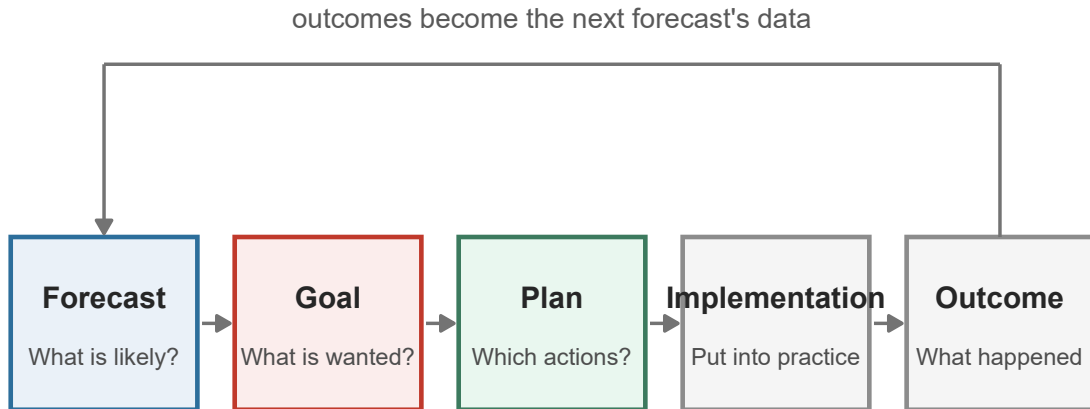
A common and expensive mistake is treating these as one activity.

Activity	Question	Character
Forecasting	What is likely to happen?	Evidence-based estimation
Goal setting	What do we want to happen?	Ambition, strategy
Planning	What should we do about it?	Action, intervention

Editing a forecast upward to match a growth target does not change what will happen - it only breaks the instrument. The loop **Forecast** → **Goal** → **Plan** → **Implementation** → **Outcome** closes: outcomes become the data behind tomorrow's forecast.

Baseline forecast 100, management goal 125, achieved outcome 118. Planning - advertising, new products, pricing - is the effort that closes the gap between the honest forecast and the desired goal.

The Forecast–Goal–Plan Loop: Figure



The loop closes on itself: implementation produces an outcome, and that outcome becomes part of the historical data the next forecast is built from.

Forecast Horizons

The horizon h is the distance between the origin T and the point predicted. The decision sets the horizon; the horizon narrows the model choice.

Horizon	Scale	Decisions
Short term	Minutes–weeks	Scheduling, inventory
Medium term	Months–2 years	Budgets, hiring
Long term	Years–decades	Infrastructure, policy

Growing Uncertainty

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Random walk

$$\text{Var}(e_{T+h}) = h\sigma^2.$$

Prediction intervals widen without bound ($\propto \sqrt{h}$).

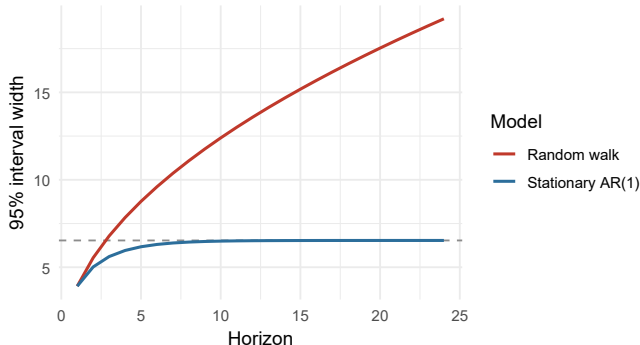
Stationary AR(1) ($|\phi| < 1$)

$$\text{Var}(e_{T+h}) = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2} \longrightarrow \frac{\sigma^2}{1 - \phi^2}.$$

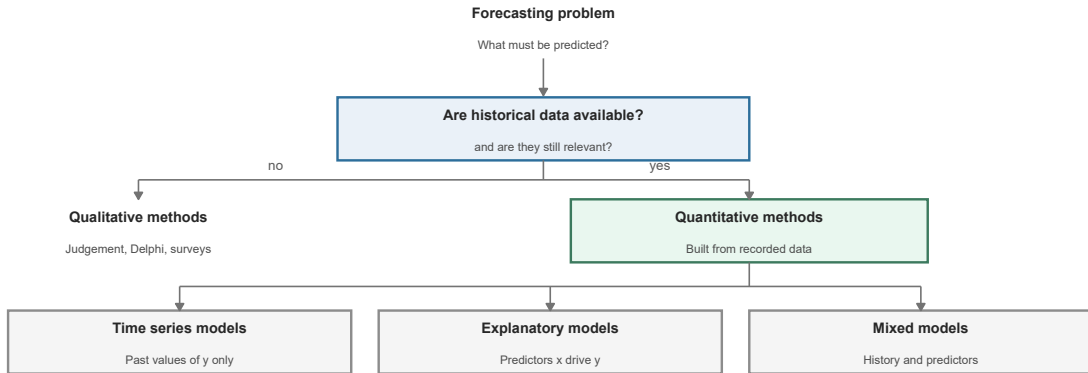
Long-horizon forecasts converge to the unconditional mean, and forecast uncertainty approaches a finite limit.

Key point: Prediction intervals widen with horizon, but only non-stationary processes continue to widen indefinitely.

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Method Families: A Decision Tree

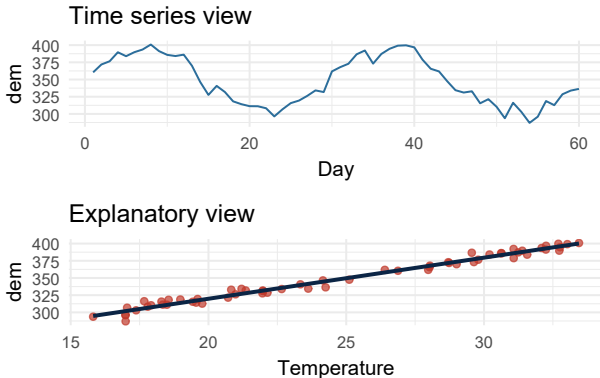


Method families divide first on a single question: are there usable historical data, and are they still relevant?

Where Forecasts Come From

- **Time series models:** history of y_t alone (exp. smoothing, ARIMA, SARIMA) - most of this course
- **Explanatory models:** outside drivers x_t (regression)
- **Mixed models:** both, respecting temporal structure

Predictors in an explanatory model must themselves be forecast - a demand forecast is only as good as the weather forecast behind it.



The Five Stages of a Forecasting Project

1. **Define the problem** - what, for whom, how far ahead?
2. **Gather information** - data plus context (was a 2020 dip a lockdown or a market shift?)
3. **Explore the data** - plot everything; trend, seasonality, missing values, breaks
4. **Build and compare models** - against benchmarks, on unseen data
5. **Evaluate and iterate** - compare forecasts with what happened, feed errors back

Forecasts Are Random Variables

Because the future is uncertain, the forecast quantity is a random variable, and a forecast is a statement about its distribution.

- **Point forecast** $\hat{y}_{T+h|T}$: the single estimate
- **Prediction interval**: shaded region expected to contain the truth with stated probability (80% or 95%)

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$$

This can only be computed once time $T + h$ has arrived - which is why forecast evaluation is harder than model fitting.

Exploring a Series

What is the sum of the series?

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Two Ways of Looking at a Series

A time series can be studied from two complementary perspectives.

Perspective	Primary question
Time domain	How do earlier observations influence later ones?
Frequency domain	Which cycles explain the observed behaviour?

The two perspectives are complementary rather than competing. They examine different characteristics of the same stochastic process, and both are important in modern time series analysis.

The Time Domain

The time domain models the current observation, y_t , using previous observations such as $y_{t-1}, y_{t-2}, \dots$. This is the foundation of autoregressive (AR), moving average (MA), ARMA and ARIMA models.

The separation between two observations measured in time units. If two observations are h periods apart, the lag is h .

The Frequency Domain

Rather than asking “*What happened yesterday?*”, the frequency domain asks “*Which cycles are present in the data, and how much variation does each explain?*”

A **period** (m) is the number of observations required to complete one cycle.

A **frequency** is

$$f = \frac{1}{m}$$

which measures the number of cycles per observation.

For example, if $m = 12$,

$$f = \frac{1}{12} \approx 0.083$$

cycles per month.

Summary Comparison

	Time domain	Frequency domain
Focus	Dependence over time	Cyclical behaviour
Main question	Does the past help predict the future?	Which oscillations are present?
Typical tools	ACF, PACF, ARIMA	Periodogram, spectral analysis
Direct forecasting	Yes	Indirect

The frequency domain does **not** introduce new information. It simply re-expresses the same n observations as a combination of sine and cosine waves. Nothing is gained or lost; only the perspective changes, making cyclical behaviour much easier to identify.

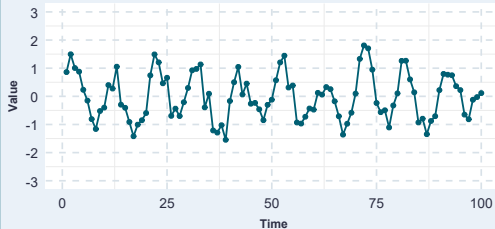
Comparing the Two Domains

Two ways of looking at a series

Neither is superior. They examine different features of the same stochastic process, and serious work usually needs both.

Primary question

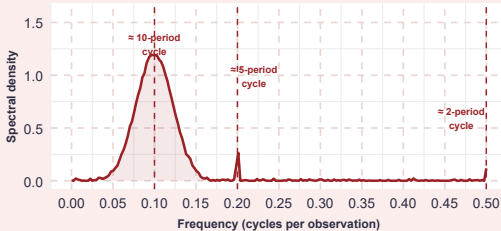
How do earlier observations influence later ones?



Tools and ideas
Autocorrelation, partial autocorrelation, ARIMA, state space,
Granger causality, impulse responses

Primary question

Which cycles explain the observed behaviour?



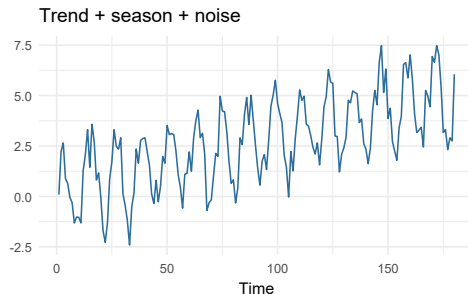
Tools and ideas
Periodogram, spectral density, coherence,
bandwidth, filtering

The Classical Decomposition

$$y_t = T_t + S_t + C_t + I_t$$

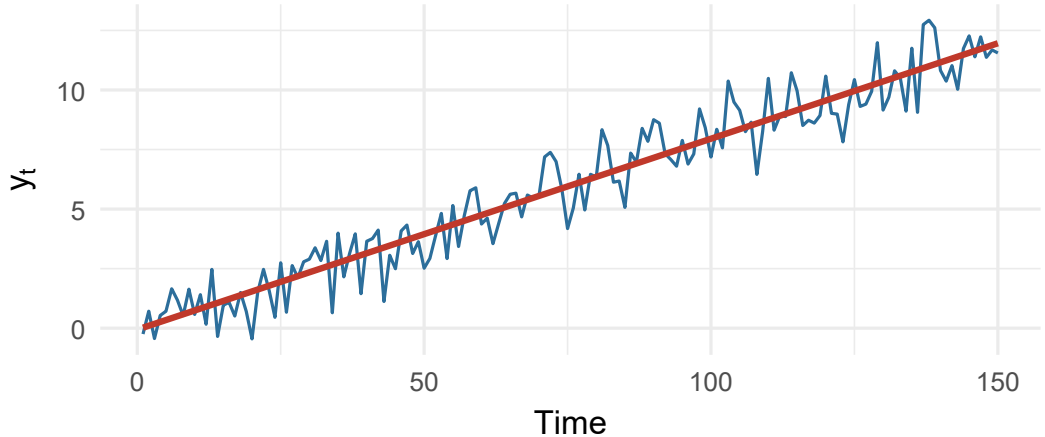
Symbol	Component
T_t	Trend, long-run direction
S_t	Seasonality, fixed repeat
C_t	Cycle, variable length
I_t	Irregular, what's left

Multiplicative form $T_t \times S_t \times C_t \times I_t$ fits growing seasonal swings; logs turn it back into the additive case.



Trend

The long-run direction of movement in a series, once short-term fluctuation is set aside.



Seasonality

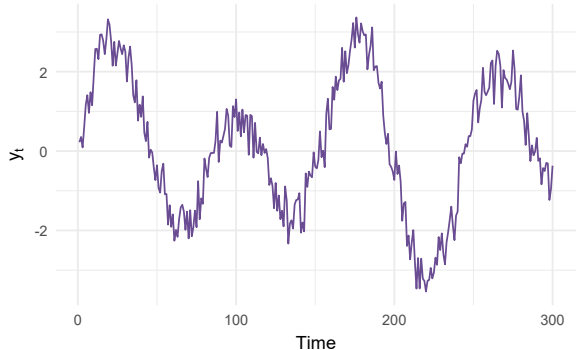
A pattern that repeats at a fixed and known interval - month of year, day of week.

The constant spacing between peaks is what separates seasonality from a cycle. A weekly pattern in daily data is seasonality with $m = 7$, not a cycle, because its length is fixed by the calendar.

Cycles

A long oscillation whose length is **not fixed**.
The business cycle is the classic example.

Seasonality	Cycle
Fixed period	Variable period
Calendar-driven	Process-driven
Highly regular	Much less regular



Two superimposed oscillations of different length - peaks no longer evenly spaced, unlike pure seasonality.

Irregular Variation

The random variation left unexplained by trend, seasonality, and cycle - usually called noise.

It turns out to be the building block from which every model in this course is assembled, not merely a residue - see the white noise section ahead.

Trend Forms and Mechanisms

Form	Characteristic	Example
Flat / Linear	Constant / constant rate	Heart rate / depreciation
Exponential	Rate $\propto$ level	Early epidemic spread
Damped	Growth slows to a ceiling	Market penetration
Polynomial	Curvature reverses	Product life cycle
With waves	Long-run + cyclical	Property markets

Mechanism	Description
Deterministic	Fixed function + noise; removing it leaves a stationary remainder
Stochastic	Accumulated random shocks; needs differencing, not detrending

Distinguishing the two is a central goal of stationarity analysis; unit root tests are designed exactly for this.

Seasonality Shapes

Shape	Characteristic	Example
Additive	Constant amplitude	Residential electricity
Two peaks	Two maxima per cycle	University enrolments
Single spike	One abrupt event	Black Friday sales
Multiplicative	Amplitude grows with level	Airline passengers
Shifting	Peak timing moves	Harvest dates
Fading	Effect weakens over time	Novelty product sales

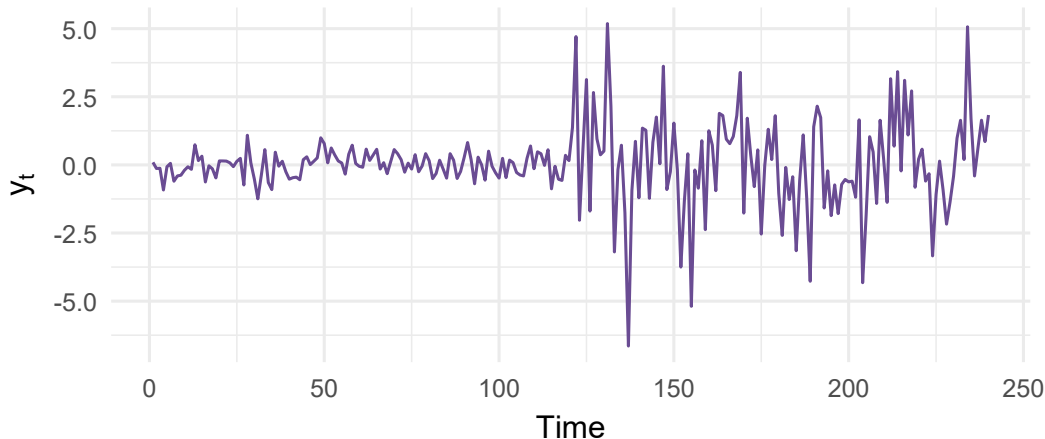
Whether seasonality is additive, multiplicative, or drifting decides whether a log transform, a seasonal difference, or both are required.

Reading a Time Plot

Before any model is fitted, a time plot should answer four questions:

1. Is the series trending?
2. Is it seasonal?
3. Is the variability stable?
4. Are there observations that do not belong?

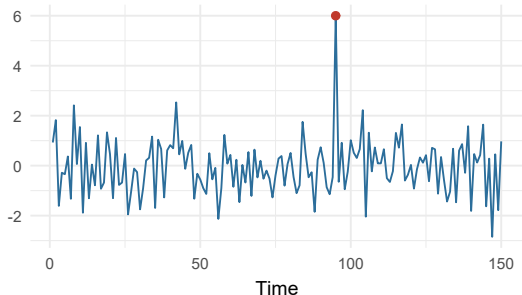
Changing Variability



Heteroscedasticity - common in financial data. Log/sqrt transforms often help; persistent, forecastable variance calls for ARCH/GARCH.

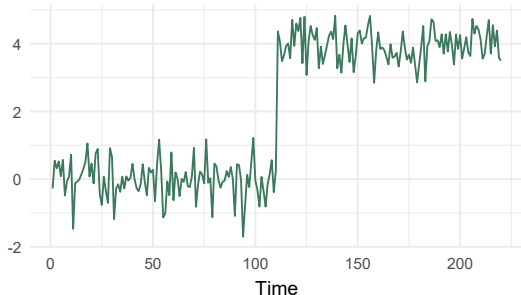
Outliers and Structural Breaks

Outlier



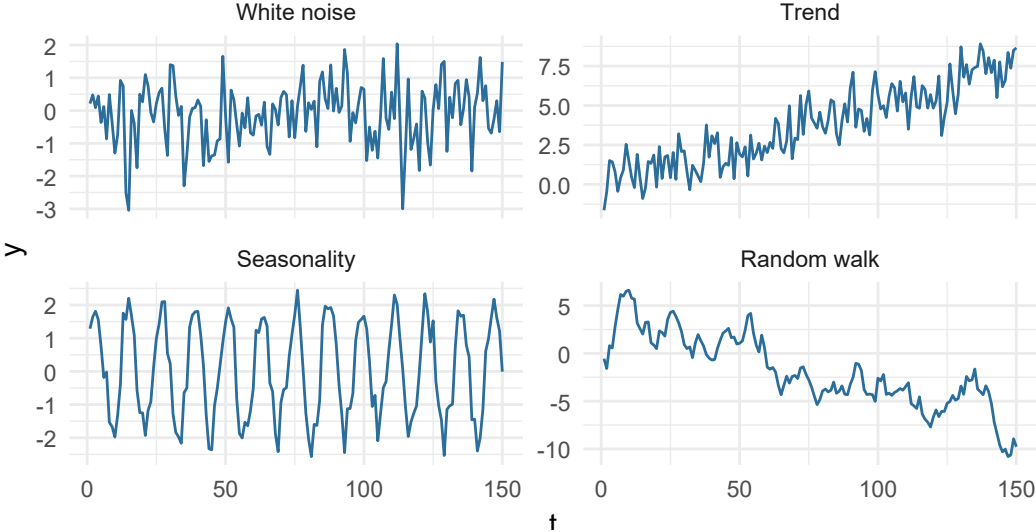
Recording errors, equipment failure, and genuine rare events all look alike - a prompt to check the source, not a licence to delete a point.

Structural break



The level shifts abruptly and does not return. One model across a break describes neither regime.

Four Archetypes



A Working Routine

Time plot → identify dominant features → describe numerically → decide on transform/differencing → **only then** fit a model.

Monthly retail sales: growth over 10 years, a December peak of fixed timing, and one pandemic year that falls sharply. Model must handle trend and seasonality together, pointing to seasonal models later in this course.

Benchmarks and Forecast Accuracy

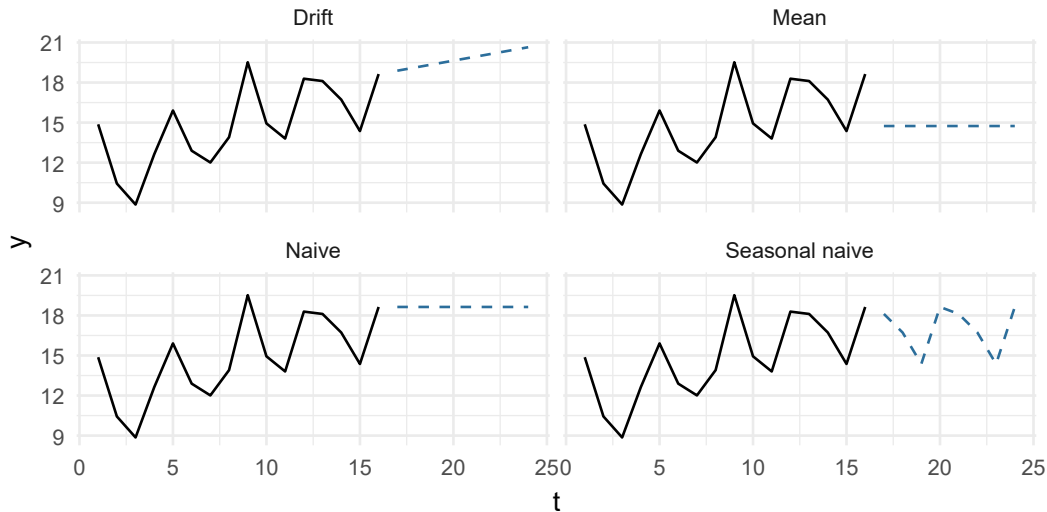
Benchmark Forecasting Methods

Any model must beat forecasts that require no modelling at all.

Method	Forecast for all h	Suited to
Mean	$\hat{y}_{T+h T} = \bar{y}$	No trend/season
Naive	$\hat{y}_{T+h T} = y_T$	Random walks, prices
Seasonal naive	$\hat{y}_{T+h T} = y_{T+h-m(k+1)}$	Strong seasonality
Drift	$\hat{y}_{T+h T} = y_T + \frac{h(y_T - y_1)}{T - 1}$	Steady trend

with $k = \lfloor (h - 1)/m \rfloor$ the complete seasons before the target - the floor is what makes $h = m$ repeat the same season one year earlier.

The Benchmarks, Applied



Residuals Are Not Forecast Errors

Residual: in-sample, $y_t - \hat{y}_t$, from a model fit on all the data.

Forecast error: out-of-sample, $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$, on data the model never saw.

A model with enough parameters drives residuals near zero and still forecasts badly - small residuals may just mean the model memorised the training sample.

Residuals are for **diagnostics**; forecast errors are for **evaluation**.

Why Cross-Validation Is Different for Time Series

- Cross-validation assesses predictive (out-of-sample) performance.
- For **independent** observations (e.g. cross-sectional survey data), random train/validation splits are fine, because the order of collection carries no statistical information.
- **Time series are different:** successive observations are correlated, and their *ordering* carries information about the data-generating process.
Future observations must never be used when estimating a forecasting model.
- Randomly splitting a time series therefore violates a fundamental principle of forecasting.

Temporal Ordering & the Fundamental Rule

Suppose $y_1, y_2, \dots, y_T$, and a forecast of y_{T+1} is required. The model may only use $\{y_1, \dots, y_T\}$.

Writing $\mathcal{T}$ for the training set and $\mathcal{V}$ for the validation set, a valid forecasting strategy satisfies, for every iteration:

$$\max(\mathcal{T}) < \min(\mathcal{V})$$

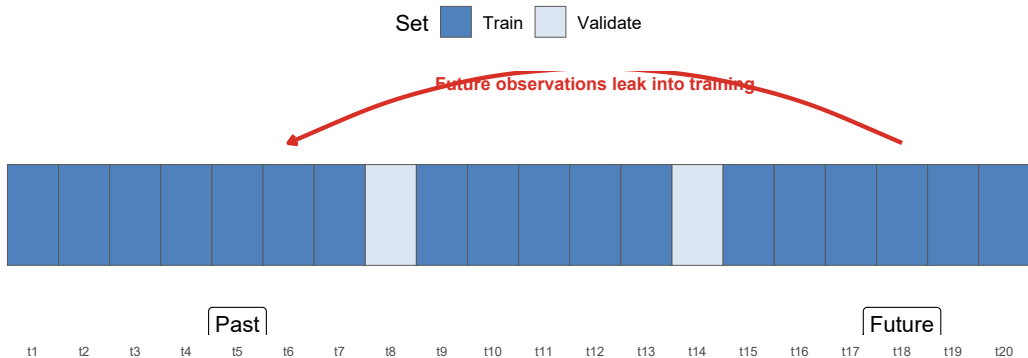
Equivalently: **every training observation precedes every validation observation.**

Temporal leakage (look-ahead bias): future observations enter model estimation. Reported accuracy then looks excellent, but is unattainable in practice - the future is unknown when real forecasts are made.

Incorrect: Standard Random Cross-Validation

Random k-Fold Cross Validation

Random sampling destroys temporal structure



- Training observations may occur after validation observation - this is the actual problem, not simply where each point happens to fall on the page.
- The model “knows” things it could not have known at forecast time.

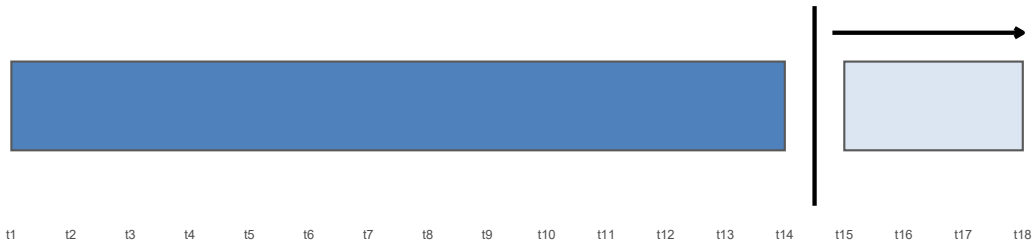
Correct Time Series Train–Validation Split

Correct Time Series Validation

Historical observations precede all validation observations

Type  Train  Validate

Forecast Origin Forecast Horizon



- The **forecast origin** is the last observation available when the forecast is made.
- For that forecast origin: everything to its left is usable for estimation; everything to its right is unseen until evaluated. (In rolling-origin evaluation the origin itself moves forward — see later.)

Training and Test Sets

- Hold out the last portion as a test set; fit on training only; compare forecasts to held-out values
- The test set should be at least as long as the largest horizon of interest
- The test set should be reserved for the final evaluation after model selection, not used repeatedly while choosing between models

Accuracy Measures

Measure	Definition	Notes
MAE	$\text{mean}(e_t)$	Scale-dependent; optimal forecast = median (under absolute loss)
RMSE	$\sqrt{\text{mean}(e_t^2)}$	Scale-dependent; optimal forecast = mean (under squared loss); penalises large errors
MAPE	$\text{mean}(100 e_t/y_t)$	Scale-free; undefined at $y_t = 0$; asymmetric
MASE	$\text{mean}(e_t)/Q$	Scale-free; where Q is the in-sample MAE of the appropriate benchmark (naive or seasonal naive)

$\text{RMSE} \geq \text{MAE}$ always. $\text{MASE} < 1$ means the method beat the naive benchmark in-sample.

Accuracy Demo Table

```
train <- window(y_all, end=c(8,12))
test  <- window(y_all, start=c(9,1))
fits <- list(Mean=meanf(train,h=24), Naive=naive(train,h=24),
  `Seasonal naive`=snaive(train,h=24),
  Drift=rwf(train,h=24,drift=TRUE))
lapply(fits, \(f) accuracy(f,test)["Test set",
  c("RMSE","MAE","MAPE","MASE")])
```

In this simulated example, seasonal naive happens to win the accuracy table; whichever benchmark wins on a given test set sets the bar that a more complex model must beat before its extra parameters are justified - this ranking is data-dependent, not a general result.

Three Validation Strategies

- **Internal Validation**
 - **Holdout** (train-test)
 - **cross-validation**
 - **bootstrap**
- **External Validation**
 - **temporal**
 - **spatial**
 - **simulated**
- **Real-world Validation**
 - **prospective**
 - **retrospective**

- **Validation Metrics**
 - **Accuracy**
 - **Precision**
 - **Recall**
 - **F1 Score**
 - **AUC-ROC**
 - **Confusion Matrix**
- **Model Interpretability**
 - **SHapley**
 - **LIME**
 - **Permutation Importance**
- **Ethical Considerations**
 - **Bias and Fairness**
 - **Transparency**
 - **Accountability**

Rolling-Origin Evaluation

A single split rests on one draw of the forecast origin. **Time series cross-validation** repeats fit-forecast-compare at many origins and pools the errors - the forecasting analogue of k -fold CV, but observations are **not exchangeable**, so folds cannot be shuffled at random.

For origins $T = T_0, T_0 + 1, \dots, n - h$: fit on $\{y_1, \dots, y_T\}$, forecast h steps, store

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T},$$

advance T by one, repeat. Averaging over every origin gives a far less noisy accuracy picture than one split, at the cost of refitting many times.

Overview of the strategies

All three strategies below preserve temporal order ($\max(\mathcal{T}) < \min(\mathcal{V})$); they differ in **how the training window evolves**:

1. **Walk-Forward Validation** - the forecast origin advances one step at a time, and the model is re-estimated at each step.
2. **Expanding Window Validation** - every historical observation is kept; the training set only grows.
3. **Sliding (Rolling) Window Validation** - the training window has a fixed length w and slides forward.

A Note on Terminology

- In much applied literature, “**walk-forward validation**” is used as an **umbrella term** for rolling the forecast origin forward and re-fitting at each step.
- Within that umbrella there are commonly two variants: an **expanding-window** walk-forward (nothing discarded) and a **sliding-window** walk-forward (fixed-size window).
- This deck follows the source convention of naming all three as parallel strategies. As formalised here, “Walk-Forward” and “Expanding Window” in fact use an **identical** growth rule (training grows by one observation per step; the first observation is always retained).
- Keep this in mind if other textbooks group the terms differently.

Walk-Forward Validation - Formulation

Training grows by one observation each iteration:

$$\mathcal{T}_i = \{y_1, \dots, y_{t+i-1}\}, \quad \mathcal{V}_i = \{y_{t+i}\}$$

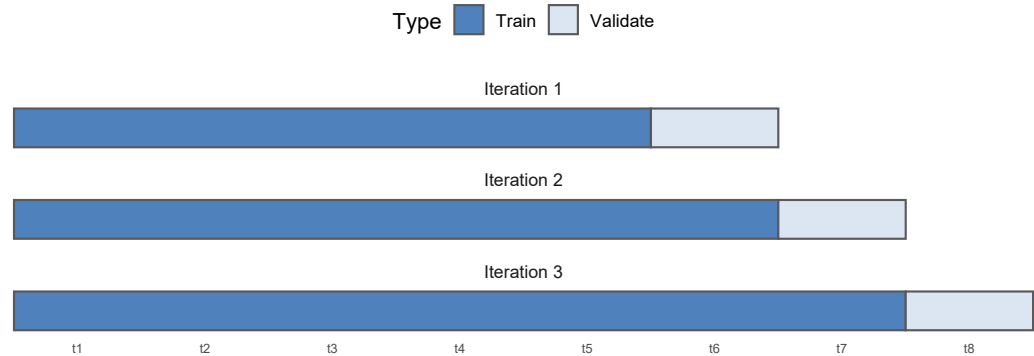
Iteration	Training	Validation
1	$y_1, \dots, y_t$	y_{t+1}
2	$y_1, \dots, y_{t+1}$	y_{t+2}
3	$y_1, \dots, y_{t+2}$	y_{t+3}

Each validation observation is folded into training once its true value is known, and the model is re-estimated before the next forecast.

Walk-Forward Validation - Illustration

Walk-Forward Validation

Training grows; the validation point moves one step ahead



Walk-Forward Validation - Advantages & Limitations

Advantages

- Closely reproduces real forecasting; prevents temporal leakage
- Every validation point is genuinely unseen; estimates improve as more data arrive

Limitations

- Requires repeated re-estimation, raising computational cost
- Typical uses: economic, electricity-demand, and epidemiological forecasting

Expanding Window Validation - Formulation

$$\mathcal{T}_i = \{y_1, \dots, y_{n_0+i-1}\}, \quad \mathcal{V}_i = \{y_{n_0+i}\}$$

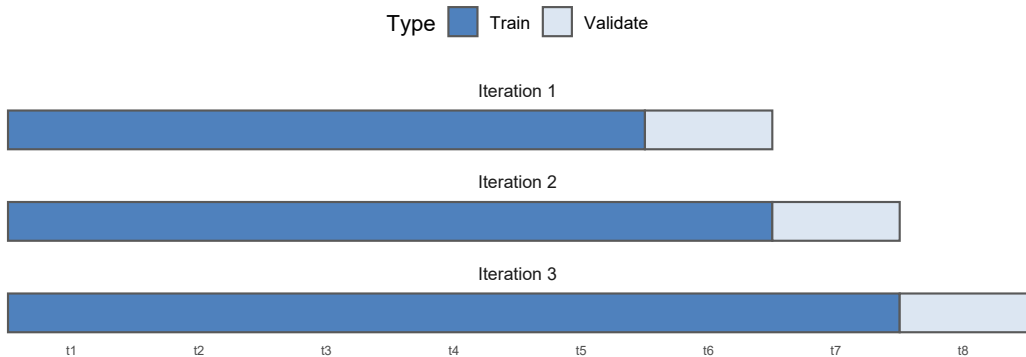
so $|\mathcal{T}_i| = n_0 + i - 1$ grows, while the start of the window (observation 1) never changes.

- The training window **always begins at the first observation**.
- Well suited when the process is relatively stable and long-run history stays informative.

Expanding Window Validation - Illustration

Expanding Window Validation

All history retained; only the right edge advances



Expanding Window Validation - Advantages & Limitations

Advantages

- Uses the complete historical record, giving stable, lower-variance parameter estimates

Limitations

- Slow to adapt after structural breaks; older data may lose relevance
- Typical uses: climatology, hydrology, national economic indicators

Sliding (Rolling) Window Validation - Formulation

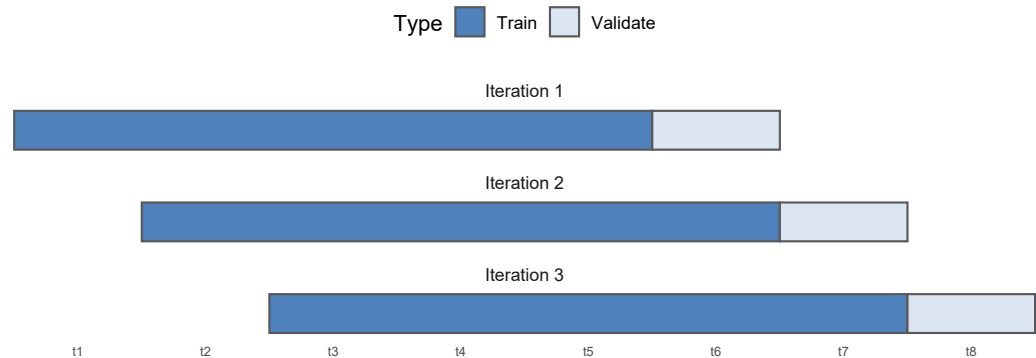
$$\mathcal{T}_i = \{y_i, \dots, y_{i+w-1}\}, \quad \mathcal{V}_i = \{y_{i+w}\}$$

Unlike the expanding window, $|\mathcal{T}_i| = w$ stays **constant**: the oldest observation is dropped as the newest one is added.

Sliding Window Validation - Illustration

Sliding (Rolling) Window Validation

Fixed-length window moves forward through time



Sliding Window Validation - Advantages & Limitations

Advantages

- Adapts rapidly to structural change; constant computational cost; suits streaming data

Limitations

- Discards potentially valuable history; more variable parameter estimates
- Window length w is a bias–variance trade-off: small w adapts fast but is noisy; large w is stable but slow to react
- Typical uses: financial markets, algorithmic trading, online recommendation systems

Implementation: tsCV()

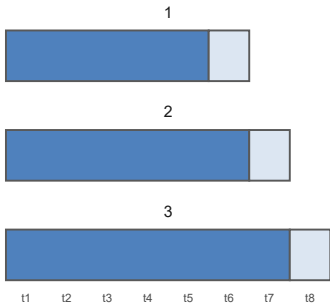
```
f_snaive <- function(x,h) forecast(snaive(x), h=h)
cv <- tsCV(y_all, f_snaive, h=12)           # expanding
rmse_h <- sqrt(colMeans(cv^2, na.rm=TRUE))

cv_slide <- tsCV(y_all, f_snaive, h=1, window=60) # sliding
rbind(
  Expanding = sqrt(mean(cv[,1]^2, na.rm=TRUE)),
  Sliding   = sqrt(mean(cv_slide^2, na.rm=TRUE)))
```

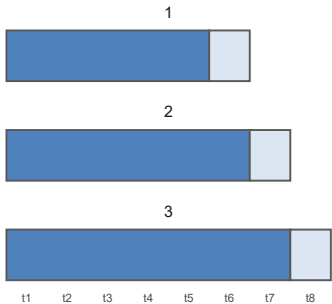
RMSE rises with horizon: one month ahead uses recent information that thirteen months ahead cannot.

Side-by-Side Comparison

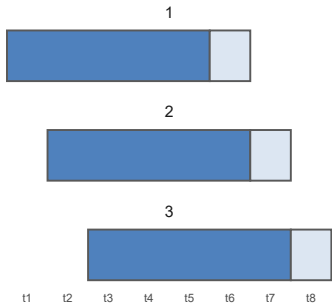
Walk-Forward



Expanding



Sliding



Statistical Comparison

Property	Walk-Forward	Expanding	Sliding
Preserves temporal order	Yes	Yes	Yes
Prevents temporal leakage	Yes	Yes	Yes
Training size increases	Yes	Yes	No
Window length fixed	No	No	Yes
Adapts to structural change	Moderate	Low	High
Uses complete history	Yes	Yes	No
Computational cost	High	Moderate	Moderate

Bias-Variance Trade-off

- **Stable process** $\Rightarrow$ older data still informative $\Rightarrow$ **Expanding Window** reduces estimation *variance*.
- **Rapidly changing process** $\Rightarrow$ older data less informative $\Rightarrow$ **Sliding Window** reduces *bias* by focusing on recent data.
- **Walk-Forward** sits between the two and most closely mimics day-to-day operational forecasting.

Choosing a Validation Strategy

Situation	Recommended Strategy
One-step-ahead, operational forecasting	Walk-Forward
Long, stable historical record (climate, demographics)	Expanding Window
Structural breaks expected	Sliding Window
Financial markets / streaming data	Sliding Window

Practical Considerations and Limits

Question	Favours
Stable process, long informative history?	Expanding window
Structural breaks plausible?	Sliding window
Forecasts updated every period?	Expanding (walk-forward)
Refitting at every origin is cheap?	Either; else, a single split

- Refitting at every origin multiplies computational cost by the number of origins
- Errors from neighbouring origins overlap and correlate - treat CV accuracy as more reliable than one split, not as independent replicates
- Answers **how a method has performed across history**, not how it will perform if the process itself is changing

Common Mistakes

Students frequently:

- Randomly shuffle observations before fitting a forecasting model
- Confuse *interpolation* with *forecasting*
- Use future observations during parameter estimation
- Compare models using only a single train/test split
- Ignore structural breaks when choosing a validation strategy

Summary

- Time series observations are **not exchangeable** - order carries information.
- Every forecast must be generated using only information available at its forecast origin.
- Forecast accuracy (MAE, RMSE, MAPE, MASE) should be compared on held-out data, never in-sample.
- **Walk-Forward**: origin advances one step at a time, re-estimating as it goes.
- **Expanding Window**: retains the complete history; the training set only grows.
- **Sliding Window**: fixed-length window; adapts to structural change.
- Choosing the right validation strategy matters as much as choosing the right model.

Exercises

1. Why is ordinary random k -fold cross-validation inappropriate for forecasting?
2. Define temporal leakage.
3. Contrast Walk-Forward, Expanding Window, and Sliding Window validation.
4. When would Sliding Window outperform Expanding Window?
5. A firm refits its model daily using only the most recent 250 observations. Which strategy is this?
6. Why must temporal ordering be preserved during validation? # Operators, Stationarity and White Noise

Why an Operator Is Worth the Trouble

Rainfall, GDP, inflation, exchange rates, share prices - all usually depend on their own recent past. Spelling out “the value two periods ago” repeatedly becomes tiresome and error-prone.

- **Lag operator** L : looks to the past
- **Lead operator** F : looks to the future (forecasting)

Neither touches the numbers - they change **which time label** a value sits under. Most notational mistakes come from losing that distinction.

A Series to Work With

$$y_t = \{3, 7, 4, 8, 5\}, \quad t = 1, \dots, 5.$$

t	1	2	3	4	5
y_t	3	7	4	8	5
Ly_t	NA	3	7	4	8
Fy_t	7	4	8	5	NA

The Lag Operator

$Ly_t = y_{t-1}$: at time t , use the observation recorded at $t - 1$.

Checking against the table: $Ly_2 = y_1 = 3$ - the value 3, recorded at $t = 1$, appears under $t = 2$. The subscript records **where a value came from**, not where it is displayed; that distinction is why the shift reads to the right.

The Lead Operator

$Fy_t = y_{t+1}$: at time t , use the observation at $t + 1$. It is the inverse of the lag operator, $F = L^{-1}$.

$Fy_2 = y_3 = 4$. Lag and lead are exact mirror images: one borrows from the past, one from the future. Lagging leaves a gap at the start; leading leaves one at the end.

Doing It in R

```
y <- c(3, 7, 4, 8, 5)
c(NA, y[1:(length(y)-1)]) # lag by hand
#> [1] NA 3 7 4 8
dplyr::lag(y, n = 1)       # same shift
#> [1] NA 3 7 4 8
dplyr::lead(y, n = 1)      # lead
#> [1] 7 4 8 5 NA
```

Warning: `stats::lag()` shifts a `ts` object's **time base** and moves it **forward** - the opposite sign convention to `dplyr::lag()`. Write `dplyr::lag()` or `stats::lag()` in full.

The Algebra of L

- **Repeated application:** $L^2 y_t = L(Ly_t) = y_{t-2}$, and in general $L^k y_t = y_{t-k}$
- **Lead as inverse:** $L(Fy_t) = Ly_{t+1} = y_t$, so $F = L^{-1}$
- **Linearity:** $L(ay_t + bx_t) = ay_{t-1} + bx_{t-1}$, just like differentiation
- **Compound operators:** $1 - \phi_1 L - \phi_2 L^2$ is well formed in its own right - not shorthand for something simpler

$$w_t = \nabla y_t = (1 - L)y_t, \quad \nabla_{12} y_t = (1 - L^{12})y_t$$

`diff()` returns a vector one element shorter rather than padding with NA - both represent $(1 - L)y_t$.

Putting the Operator to Work

Take $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$. Moving lag terms across:

$$y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} = \varepsilon_t \iff (1 - \phi_1 L - \phi_2 L^2)y_t = \varepsilon_t.$$

Stack enough of these and you reach the general ARIMA form, $\phi(L)y_t = \theta(L)\varepsilon_t$ - writing a model in L makes its structure visible at a glance.

Common mistakes: forgetting the subscript records **origin** not display column; losing track of which end goes missing (lag: start, lead: end); treating $\phi(L)$ as a number rather than an operator; reading $L^2 y_t$ as $(y_{t-1})^2$; confusing lagging with differencing.

Weak and Strict Stationarity

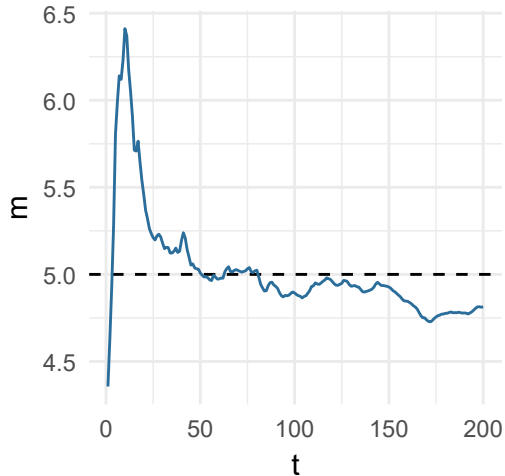
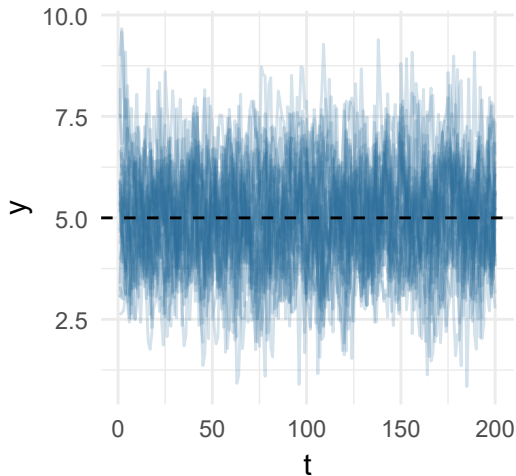
For all t, h : (1) $E(y_t) = \mu$ constant; (2) $\text{Var}(y_t) = \gamma(0) < \infty$ constant; (3) $\text{Cov}(y_t, y_{t+h}) = \gamma(h)$ depends on h only.

Joint distribution of $(y_{t_1}, \dots, y_{t_k})$ equals that of $(y_{t_1+h}, \dots, y_{t_k+h})$ for every choice of times and shift h .

Strict + finite variance $\Rightarrow$ weak, but not conversely - GARCH is weak-stationary yet has time-varying conditional variance.

Why the assumption matters: $\rho(h)$ is only well defined if $\gamma(h)$ doesn't depend on t ; $\mu, \gamma(0), \gamma(h)$ are estimated from **time averages over one realisation**, legitimate only if stationary and ergodic; and a forecast projects observed structure forward, which needs that structure to still hold tomorrow.

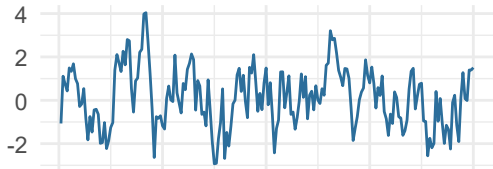
Ergodicity



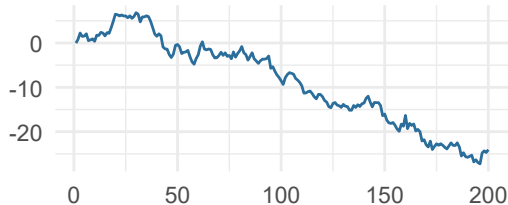
Ergodicity: time averages along one realisation converge to the same value as ensemble averages

Violations of Stationarity

Stationary AR(1)

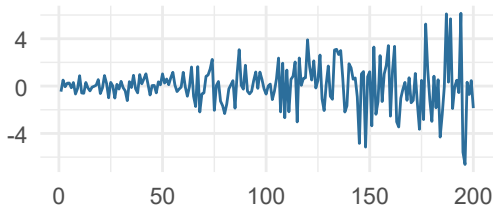


Random walk



y

Variance not constant



t

Restoring Stationarity

Symptom	Remedy
Variance grows with level	Log / Box–Cox transform
Deterministic trend	Fit and subtract the trend
Stochastic trend, unit root	First differencing
Non-fading seasonality	Seasonal differencing at lag m
Structural break	Split the sample / add an indicator

Transform before differencing - stabilising variance often reduces how much differencing is needed.
ADF (null: unit root) and KPSS (null: stationary) are usefully read together.

White Noise

$\varepsilon_t \sim WN(\mu, \sigma^2)$: uncorrelated observations, constant mean and variance. Mean is zero unless stated otherwise.

$$\rho(h) = \begin{cases} 1, & h = 0 \\ 0, & h \neq 0 \end{cases}$$

White noise is not “an interesting model of anything” - it is the **standard against which fitted models are judged**. If residuals are indistinguishable from white noise, the model has captured all linear structure.

Weak vs Strict White Noise

- **Weak white noise:** only requires **uncorrelated** observations
- **Strict white noise:** requires **independent and identically distributed** observations - much stronger

A GARCH process is weak white noise (zero autocorrelation at every lag $\neq 0$) yet plainly not independent: today's large value makes tomorrow's more likely through the conditional variance.

"Unpredictable" for white noise means **linearly** unpredictable beyond the mean. Whether a non-linear function of the past helps is separate - for volatility, the answer is yes.

Linear Models for the Conditional Mean

From White Noise to Dependence

A weighted combination of neighbouring white noise terms is not itself white noise:

$$y_t = \frac{1}{3}(\varepsilon_{t-1} + \varepsilon_t + \varepsilon_{t+1}).$$

This is **filtering**: $y_t = \sum_j a_j \varepsilon_{t-j}$. The example is two-sided ($j = -1, 0, 1$) - fine for smoothing after the fact, useless for forecasting. MA models restrict $j \geq 0$: current and past shocks only.

The MA(q) Model

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

Driven by q lagged **shocks**, not lagged observations. An MA(q) is stationary for **any** coefficient values - a finite weighted sum of white noise with fixed weights. Invertibility is the separate condition needed for estimation.

Properties of MA(1)

For $y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$:

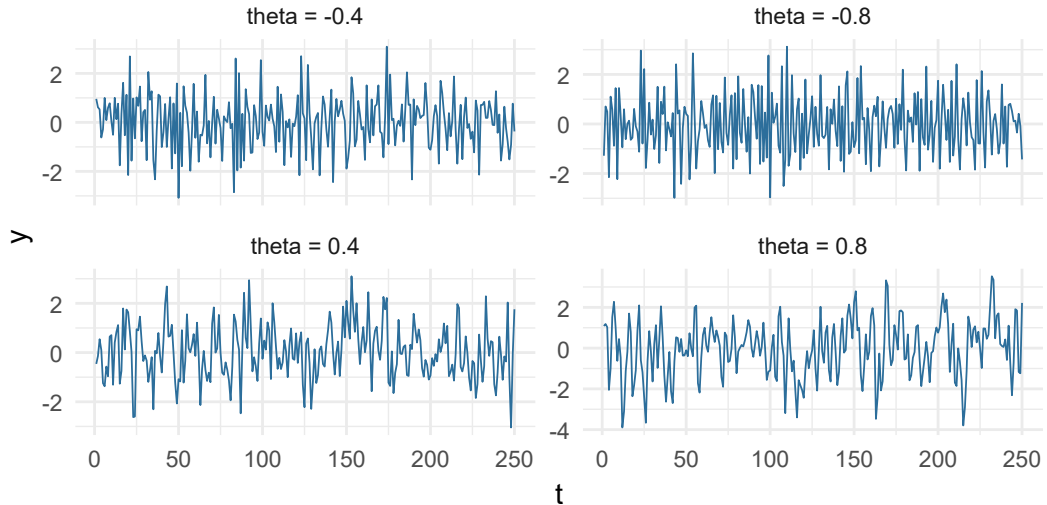
$$E(y_t) = \mu, \quad \text{Var}(y_t) = (1 + \theta^2)\sigma^2, \quad \text{Cov}(y_t, y_{t-1}) = \theta\sigma^2,$$

$$\rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(h) = 0 \text{ for } h > 1.$$

Short memory - dependence vanishes completely beyond one lag; that sharp cutoff is the identification signature.

$$y_t = 5 + \varepsilon_t + 0.6\varepsilon_{t-1}, \quad \varepsilon_t \sim WN(0, 9): \quad E(y_t) = 5, \quad \text{Var}(y_t) = (1 + .36)(9) = 12.24, \\ \text{Cov}(y_t, y_{t-1}) = 0.6 \times 9 = 5.4, \quad \rho(1) \approx 0.44.$$

MA(1) Processes: Figure



Autoregressive Processes

If a river runs high today, it will probably still run high tomorrow - temperature, unemployment, inflation all behave this way.

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

The AR(1) Process

$$y_t = c + \phi y_{t-1} + \varepsilon_t$$

ϕ	Behaviour
$\phi = 0$	White noise
$0 < \phi < 1$	Positive persistence
$-1 < \phi < 0$	Alternation
$\phi = 1$	Random walk (unit root)
$ \phi > 1$	Explosive

AR(1): Mean and Variance

Taking expectations of $y_t = c + \phi y_{t-1} + \varepsilon_t$, stationarity gives $\mu = c + \phi\mu$, so

$$\mu = \frac{c}{1 - \phi}, \quad |\phi| < 1.$$

Subtracting the mean: $y_t - \mu = \phi(y_{t-1} - \mu) + \varepsilon_t$. Since $y_{t-1} \perp \varepsilon_t$,

$$\text{Var}(y_t) = \phi^2 \text{Var}(y_t) + \sigma^2 \implies \text{Var}(y_t) = \frac{\sigma^2}{1 - \phi^2}.$$

Outside $|\phi| < 1$ this is not merely inconvenient - it is meaningless (negative variance).

AR(1): Autocorrelation Derivation

Multiplying $y_t - \mu = \phi(y_{t-1} - \mu) + \varepsilon_t$ by $(y_{t-h} - \mu)$ and taking expectations, the innovation term drops out (ε_t uncorrelated with anything before t):

$$\gamma(h) = \phi \gamma(h-1), \quad h \geq 1.$$

Solving this first-order difference equation, $\gamma(h) = \phi^h \gamma(0)$, so

$$\rho(h) = \phi^{|h|}, \quad h = 0, \pm 1, \pm 2, \dots$$

Positive ϕ decays smoothly; negative ϕ decays while alternating in sign; $|\phi|$ near 1 decays very slowly.

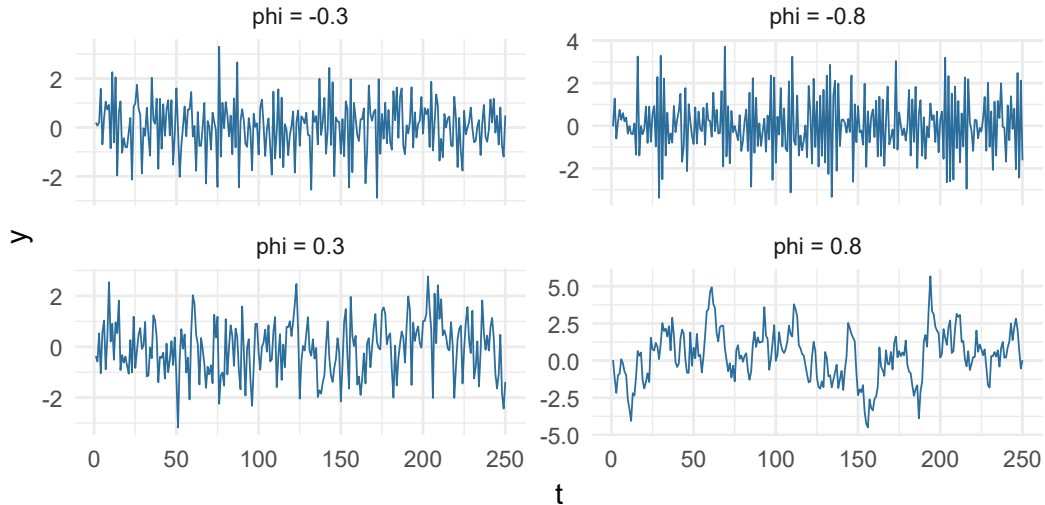
AR(1): Stationarity and Causality

In operator form: $(1 - \phi L)y_t = c + \varepsilon_t$. Stationarity requires the root of $1 - \phi z = 0$, i.e. $z = 1/\phi$, to lie **outside the unit circle**.

This is strictly a condition of **causality**. When $|\phi| > 1$, a stationary solution exists but expresses y_t in terms of *future* innovations - useless for forecasting, not identifiable from data. Every text and package adopts the causal solution.

$y_t = 3 + 0.6y_{t-1} + \varepsilon_t$, $\varepsilon_t \sim WN(0, 4)$: $\mu = 3/(1 - 0.6) = 7.5$, $\text{Var}(y_t) = 4/(1 - 0.36) = 6.25$.
Since $|0.6| < 1$, stationary.

AR(1) Processes: Figure



Autocovariance and Autocorrelation

$\gamma(h) = \text{Cov}(y_t, y_{t-h})$; for a stationary process $\gamma(0) = \text{Var}(y_t)$, $\gamma(h) = \gamma(-h)$.

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}, \quad -1 \leq \rho(h) \leq 1, \quad \rho(0) = 1.$$

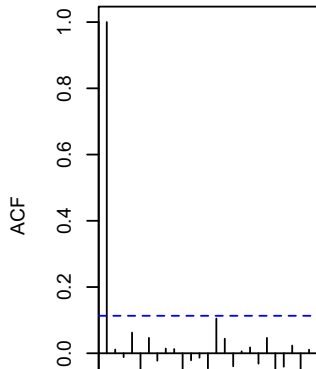
Symmetry: $\gamma(-h) = \text{Cov}(y_t, y_{t+h}) = \text{Cov}(y_{t+h}, y_t) = \gamma(h)$, using stationarity to relabel the origin.

Bound: Cauchy–Schwarz gives $|\text{Cov}(y_t, y_{t-h})| \leq \sqrt{\text{Var}(y_t) \text{Var}(y_{t-h})} = \gamma(0)$, so $|\rho(h)| \leq 1$, with equality only when y_{t-h} is an exact linear function of y_t .

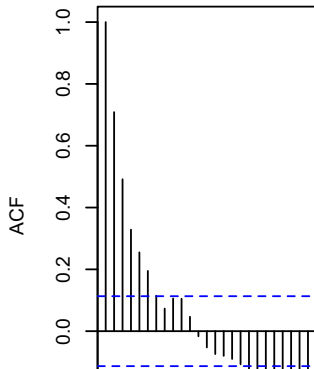
Sample ACFs: Three Processes

```
par(mfrow=c(1,3))  
acf(rnorm(300), main="White noise")  
acf(arima.sim(list(ar=.7),300), main="AR(1)")  
acf(arima.sim(list(ma=.7),300), main="MA(1)")
```

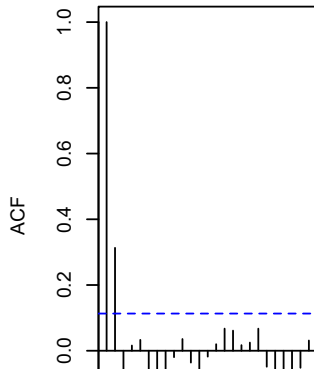
White noise



AR(1)



MA(1)



Partial Autocorrelation

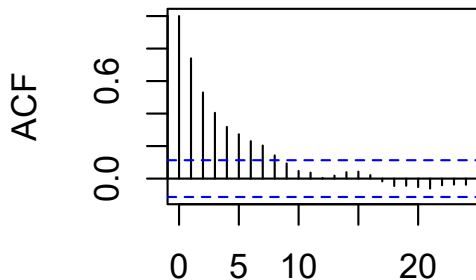
ϕ_{hh} : the correlation between y_t and y_{t-h} once the linear effect of $y_{t-1}, \dots, y_{t-h+1}$ is removed.

For AR(1): y_t depends directly only on y_{t-1} , so $\phi_{11} = \phi$ and $\phi_{hh} = 0$ for $h > 1$ - a sharp cutoff. For MA(1), the PACF decays gradually instead, the mirror image of its ACF.

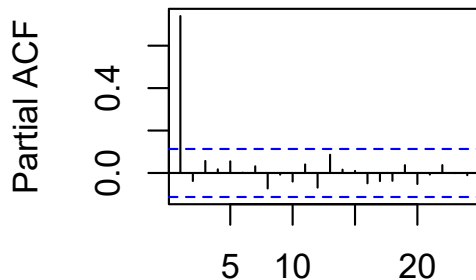
ACF and PACF of an AR(1)

```
s <- arima.sim(list(ar=.7), 300)
par(mfrow=c(1,2))
acf(s, main="ACF"); pacf(s, main="PACF")
```

ACF



PACF



Identification Table

Process	ACF	PACF
White noise	No significant spikes	No significant spikes
$AR(p)$	Decays gradually	Cuts off after lag p
$MA(q)$	Cuts off after lag q	Decays gradually
$ARMA(p, q)$	Decays gradually	Decays gradually

A guide, not a rule - sample ACFs are noisy; final choice also rests on information criteria and residual diagnostics.

PACF cuts off after lag one ($\hat{\phi}_{11} = 0.71$), later lags near zero; ACF decays gradually $\Rightarrow AR(1)$, $\hat{\phi} \approx 0.71$.

ARMA Models

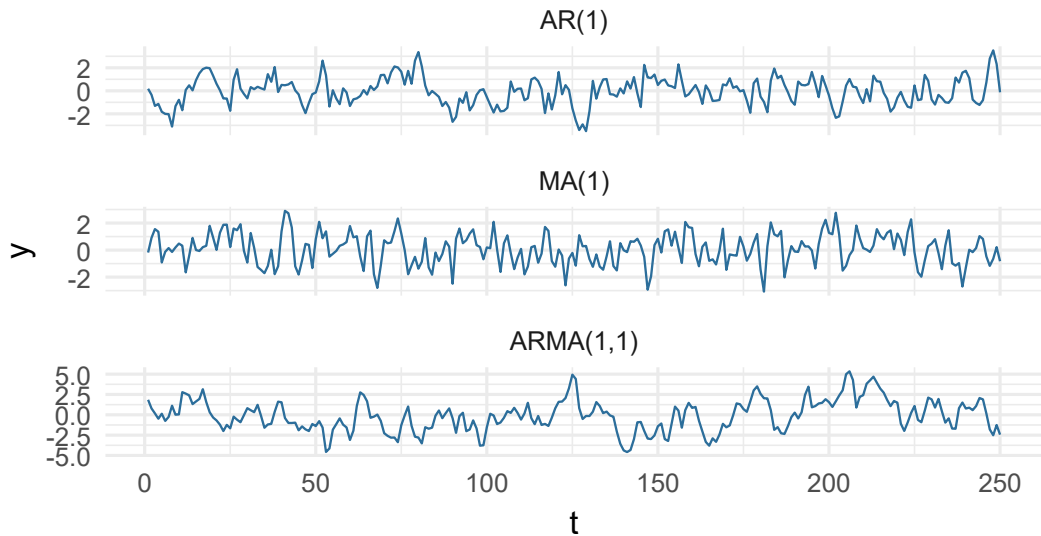
Electricity demand persists day to day, but a cold snap also leaves a shock that fades after a few periods - ARMA allows both.

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$$

or $\phi(L)y_t = c + \theta(L)\varepsilon_t$.

Simplest useful case: ARMA(1,1), $y_t = c + \phi y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}$.

AR/MA/ARMA Compared



Stationarity and Invertibility

- **AR part:** roots of $\phi(z) = 0$ outside the unit circle (as for AR(1))
- **MA part must be invertible:** roots of $\theta(z) = 0$ outside the unit circle, so it can be rewritten as an infinite autoregression

Without invertibility, two different parameter sets give the same autocorrelation function - for MA(1), θ and $1/\theta$ share an ACF, and only the value inside the unit circle is retained.

Estimation is by **maximum likelihood**, not OLS - $\varepsilon_{t-1}, \dots, \varepsilon_{t-q}$ are unobserved, so there is no design matrix. The likelihood is evaluated recursively via a state-space form and the Kalman filter; invertibility keeps that reconstruction stable. Pure AR can use least squares, but `Arima()` still reports MLE by default.

Non-Stationarity and Its Symptoms

A great many series break the constant-mean/variance/dependence assumption through a persistent trend - fitted ARMA results simply do not hold, so the series is transformed first.

$$\nabla y_t = y_t - y_{t-1} = (1 - L)y_t, \quad \nabla^2 y_t = (1 - L)^2 y_t.$$

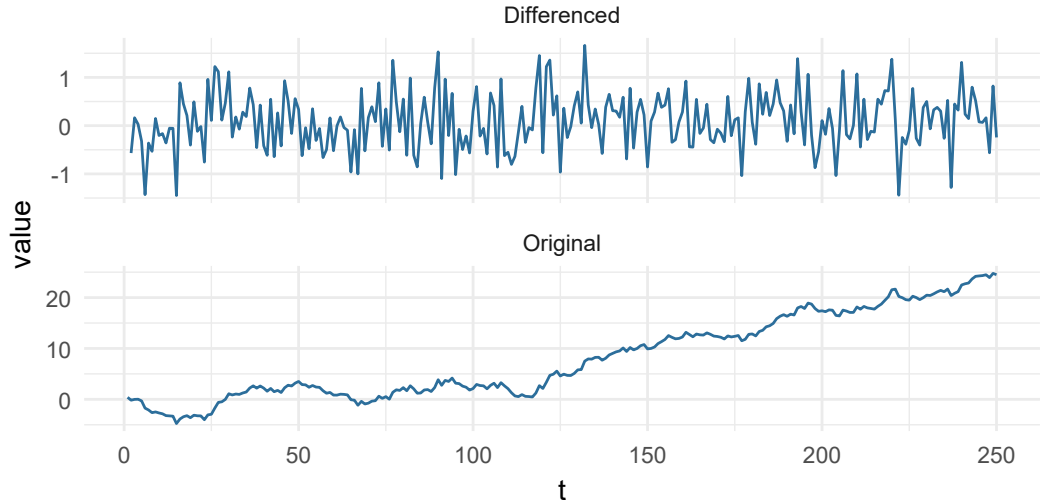
The Difference Operator: Why It Works

For $y_t = 5 + 0.4t + \varepsilon_t$:

$$\nabla y_t = (5 + 0.4t + \varepsilon_t) - (5 + 0.4(t - 1) + \varepsilon_{t-1}) = 0.4 + \varepsilon_t - \varepsilon_{t-1}.$$

The trend is gone, leaving a constant and noise. For a random walk $y_t = y_{t-1} + \varepsilon_t$: $\nabla y_t = \varepsilon_t$ exactly.

Differencing: Figure



Over-Differencing

Differencing a deterministic trend leaves $\varepsilon_t - \varepsilon_{t-1}$: an MA(1) with $\theta = -1$, exactly the non-invertible boundary.

An over-differenced series shows $\rho(1) \approx -0.5$ and an **inflated** sample variance:

$$\rho(1) = \frac{-1}{1 + (-1)^2} = -0.5.$$

If differencing makes variance go **up** rather than down, you have gone one step too far. Almost no series needs $d > 2$.

The ARIMA(p, d, q) Model

Apply d differences to y_t , then fit ARMA(p, q):

$$\phi(L)(1 - L)^d y_t = c + \theta(L)\varepsilon_t.$$

ARIMA Special Cases; Choosing d

Specification	Equivalent to
ARIMA(1,0,0) / (0,0,1)	AR(1) / MA(1)
ARIMA(1,0,1)	ARMA(1,1)
ARIMA(0,1,0)	Random walk (with drift if $c \neq 0$)
ARIMA(1,1,1)	One difference, then ARMA(1,1)

AIC/BIC compare models on the **same** data, so they may choose p, q with d fixed - but **not** d itself: ARIMA($p, 1, q$) fits $n - 1$ differenced values, ARIMA($p, 2, q$) fits $n - 2$, so likelihoods are incomparable. Choose d from plots and unit-root tests first.

From ARMA(1,1) to ARIMA(1,1,1) (1/2)

Write $w_t = y_t - y_{t-1}$, model as ARMA(1,1):

$$w_t = c + \phi w_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

Substituting $w_t = y_t - y_{t-1}$ and $w_{t-1} = y_{t-1} - y_{t-2}$:

$$y_t - y_{t-1} = c + \phi(y_{t-1} - y_{t-2}) + \varepsilon_t + \theta \varepsilon_{t-1}.$$

From ARMA(1,1) to ARIMA(1,1,1) (2/2)

Expanding the bracket and moving $-y_{t-1}$ across:

$$y_t = c + (1 + \phi)y_{t-1} - \phi y_{t-2} + \varepsilon_t + \theta \varepsilon_{t-1}$$

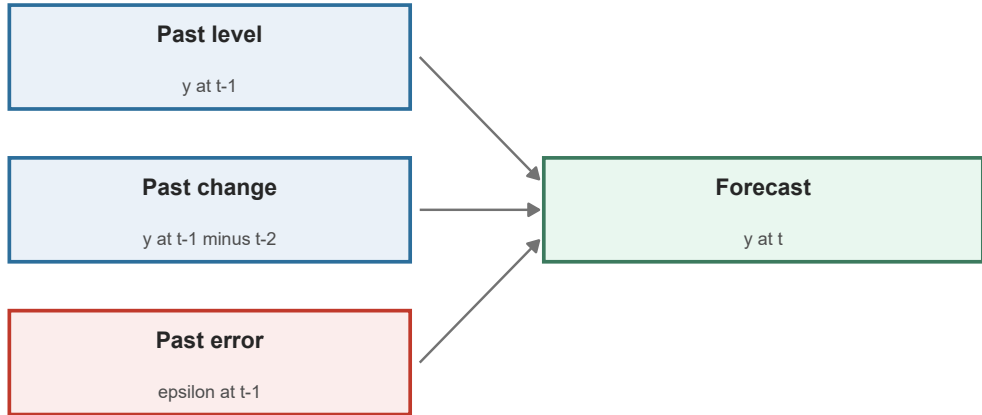
The coefficient on y_{t-1} is $1 + \phi$, not ϕ , and $-\phi y_{t-2}$ appears - both are artefacts of writing a differenced model on the original scale, not evidence of strong lag-2 dependence.

Reading the ARIMA(1,1,1) Equation

Term	Contributes
c	Constant; after differencing, drift
$(1 + \phi)y_{t-1}$	Previous level, adjusted for momentum
$-\phi y_{t-2}$	Correction turning levels into changes
ε_t	Current shock - unknowable in advance
$\theta\varepsilon_{t-1}$	Previous forecast error, fed back

$c = 1, \phi = 0.6, \theta = 0.3, y_{t-1} = 120, y_{t-2} = 115, \varepsilon_{t-1} = 2$. Replacing ε_t by 0:
 $\hat{y}_t = 1 + 1.6(120) - 0.6(115) + 0.3(2) = 124.6$ - constant 1, momentum 3, error feedback 0.6.

Three Inputs to an ARIMA Forecast



Past level, past change, and past error each feed the forecast - the third input is exactly what a pure autoregression lacks.

Fitting ARIMA in R

```
fit <- Arima(y_int, order = c(1, 1, 1))  
fit  
autoplot(forecast(fit, h = 24))
```

Point forecasts settle onto a constant level (future change decays to zero); prediction intervals keep widening (roughly $\propto \sqrt{h}$) because future innovations are unpredictable and errors accumulate.

Residual Diagnostics

If the model captured all linear structure, residuals should look like white noise. Checking lags one at a time risks false alarms by chance - a **portmanteau test** asks the joint question.

$$Q_{LB} = n(n+2) \sum_{k=1}^H \frac{\hat{\rho}_k^2}{n-k}$$

Approximately χ_{H-K}^2 under H_0 : residuals are white noise, where K = number of estimated ARMA parameters.

The K correction matters - ignoring it makes the test too conservative. Common choice: $H = 10$ (non-seasonal), $H = 2m$ (seasonal). Small p-value: evidence against the model; large p-value is the outcome you want, but absence of evidence, not proof of adequacy.

Residual Diagnostics: Figure

```
checkresiduals(fit_arima)
```

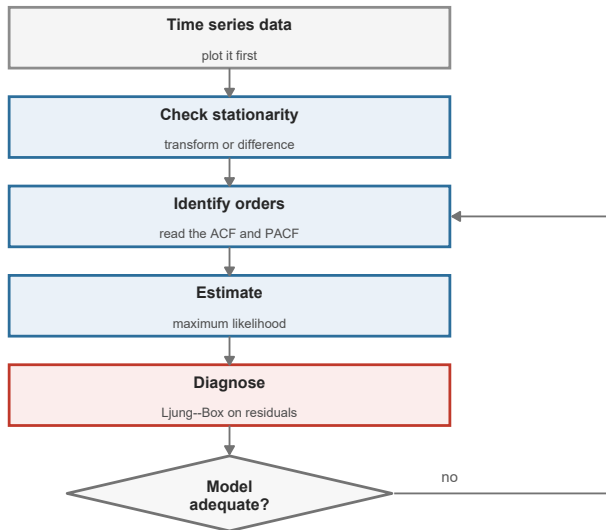
Three checks at once: residual plot (no remaining trend/season/heteroscedasticity), ACF (no significant spikes), histogram (normality of residuals, bearing on interval validity).

The Box–Jenkins Workflow

1. Plot the data
2. Difference until approximately stationary
3. Read ACF/PACF to identify candidate orders
4. Estimate by maximum likelihood
5. Diagnose residuals against the white-noise standard
6. **Forecast** - only if diagnostics pass; otherwise return to step 3

A model surviving the loop must still beat the benchmarks on the test set before it earns its place.

The Box–Jenkins Loop: Figure



Seasonal Series and SARIMA

Seasonality surviving ordinary differencing needs its own treatment.

$$\nabla_m y_t = y_t - y_{t-m} = (1 - L^m)y_t$$

where m = observations per season ($m = 12$ monthly, $m = 4$ quarterly).

Trend + season needs both: $(1 - L)(1 - L^m)y_t$.

The SARIMA Model

$$\Phi(L^m)\phi(L)(1 - L^m)^D(1 - L)^d y_t = \Theta(L^m)\theta(L)\varepsilon_t.$$

The **airline model**, SARIMA(0,1,1)(0,1,1)₁₂ - no AR terms at all - remains a remarkably strong monthly default, named for the Box–Jenkins airline passenger series. Seasonal differencing costs m observations, so it matters when the series is short.

SARIMA Forecast: Figure

```
fit <- Arima(seasonal_series, order=c(1,1,1),  
            seasonal=c(0,1,1))  
autoplot(forecast(fit, h = 36))
```

The seasonal shape carries forward while intervals widen. Compare the estimated seasonal MA coefficient with -1 : values near the boundary warn that the seasonal difference was not needed.

Conditional Variance Models

Why a Second Equation Is Needed

An ARMA model says where a series is going next; the spread was assumed fixed - one σ^2 for the whole sample. For financial returns this understates risk exactly when risk is highest, and overstates it when markets are quiet.

$$r_t = \ln p_t - \ln p_{t-1} = (1 - L) \ln p_t$$

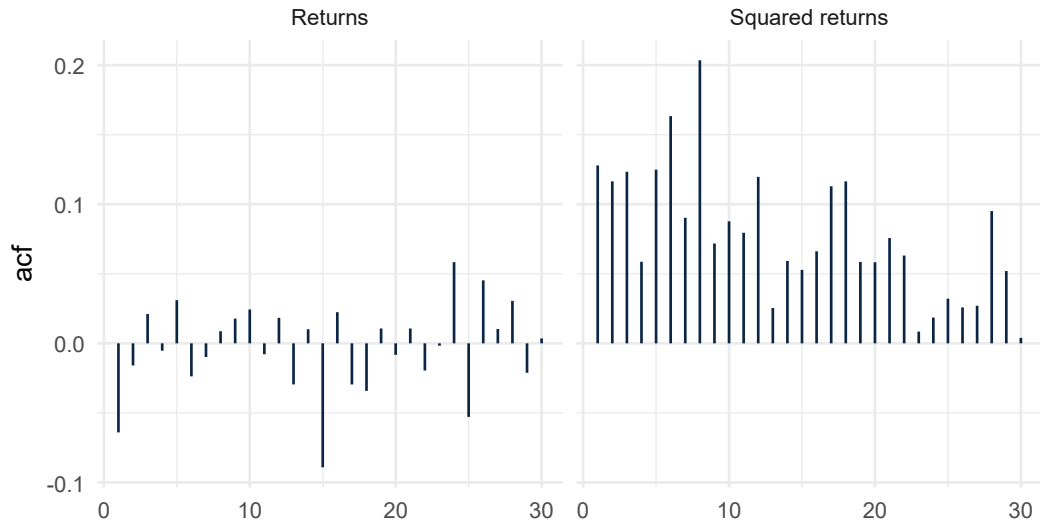
Log returns add across time (no compounding); simple returns add across assets - mixing the two is a common source of small persistent errors.

The Stylised Facts

Fact	Observation
Near-zero autocorrelation	ACF of r_t shows almost nothing
Strong autocorrelation in squares	ACF of r_t^2 decays slowly
Volatility clustering	Large moves follow large moves
Heavy tails	Far more extremes than Normal allows
Leverage effect	Falls raise future volatility more than equal rises

A series can be **uncorrelated** and still highly **predictable in magnitude** - the whole motivation for this chapter.

Stylised Facts: Figure



Testing for ARCH Effects

Regress $\hat{\varepsilon}_t^2$ on a constant and its own first q lags:

$$\hat{\varepsilon}_t^2 = a_0 + a_1 \hat{\varepsilon}_{t-1}^2 + \cdots + a_q \hat{\varepsilon}_{t-q}^2 + u_t.$$

Under H_0 (no ARCH), $nR^2 \sim \chi_q^2$ approximately.

If yesterday's squared shock helps predict today's squared shock, the variance is not constant. Ljung–Box on the **squared** residuals gives the same signal.

The ARCH Model

$$y_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \text{ iid}, \quad E(z_t) = 0, \quad \text{Var}(z_t) = 1$$
$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2, \quad \alpha_0 > 0, \quad \alpha_i \geq 0.$$

Two noise terms doing different jobs: z_t is **strict** white noise; ε_t is only **weak** white noise - uncorrelated but with a partially predictable magnitude.

Properties of ARCH(1)

For $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$:

- $E(\varepsilon_t | \mathcal{I}_{t-1}) = 0$ - the mean equation is undisturbed
- Innovations are uncorrelated at every lag $\neq 0$ - why the ACF of returns shows nothing
- Long-run variance: $\bar{\sigma}^2 = \alpha_0 + \alpha_1 \bar{\sigma}^2 \Rightarrow \bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1}$
- A mixture of Normals is not Normal - heavy tails emerge even with Normal z_t

$\hat{\alpha}_0 = 0.04, \hat{\alpha}_1 = 0.45$: $\bar{\sigma}^2 = 0.04/0.55 \approx 0.0727$. A shock four times usual size gives $\sigma_t^2 = 0.04 + 0.45(1.164) \approx 0.564$ - nearly $8\times$ resting level, but for one period only.

The GARCH Model

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

$\alpha_0 > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$.

$p = q = 1$ is almost always enough. A useful default: fit GARCH(1,1) first; treat any apparent need for more as a sign of a misspecified mean or a structural break.

GARCH(1,1) as an ARCH Process with Infinite Memory

Unrolling the recursion for $\sigma_{t-1}^2, \sigma_{t-2}^2, \dots$:

$$\sigma_t^2 = \frac{\alpha_0}{1 - \beta_1} + \alpha_1 (\varepsilon_{t-1}^2 + \beta_1 \varepsilon_{t-2}^2 + \beta_1^2 \varepsilon_{t-3}^2 + \dots).$$

Three parameters generate an unbounded sequence of geometrically declining ARCH weights - exactly the ARMA-over-AR bargain, applied to variance.

Reading a Fitted GARCH(1,1)

Parameter	Governs	Reading
α_0	Level	Sets $\bar{\sigma}^2$ jointly with the rest
α_1	Reaction	How sharply volatility jumps on one large day
β_1	Memory	How long that jump takes to fade
$\alpha_1 + \beta_1$	Persistence	Overall speed of reversion

Two markets can share persistence and look different: small α_1 /large β_1 = smooth path; large α_1 /small β_1 = spiky path.

Long-Run Variance, Persistence, Half Life

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \alpha_1 + \beta_1 < 1 \text{ (stationarity).}$$

$$h_{1/2} = \frac{\ln 0.5}{\ln(\alpha_1 + \beta_1)}$$

Persistence	Half life
0.80	≈ 3 periods
0.95	≈ 14 periods
0.99	≈ 69 periods (typical daily equities)

$\hat{\alpha}_0 = 0.012, \hat{\alpha}_1 = 0.085, \hat{\beta}_1 = 0.905$: persistence 0.990, $\bar{\sigma}^2 = 1.20$ (SD $\approx 1.10\%$ /day, $\approx 17.4\%$ annualised), half life ≈ 69 days - reacts modestly, forgets slowly.

Forecasting Volatility

One step ahead requires no forecasting - every term is known at T :

$$\sigma_{T+1}^2 = \alpha_0 + \alpha_1 \varepsilon_T^2 + \beta_1 \sigma_T^2.$$

Beyond one step, the recursion solves (same form as AR(1)):

$$\hat{\sigma}_{T+h}^2 = \bar{\sigma}^2 + (\alpha_1 + \beta_1)^{h-1}(\sigma_{T+1}^2 - \bar{\sigma}^2)$$

Volatility forecasts revert geometrically to the long-run level - from above after a crisis, from below in a quiet spell. Since returns are uncorrelated, the variance of a cumulative h -period return is the **sum** $\sum_{k=1}^h \hat{\sigma}_{T+k}^2$ - the square-root-of-time rule is the special case that pretends volatility never reverts.

GARCH Fitted Bands

```
library(rugarch)
spec <- ugarchspec(
  variance.model = list(model="sGARCH", garchOrder=c(1,1)),
  mean.model = list(armaOrder=c(0,0)),
  distribution.model = "std")
fit <- ugarchfit(spec, data = returns)
coef(fit)
```

The fitted $\pm 2\hat{\sigma}_t$ band tracks the series: wide where returns are wide, narrow where calm - a constant-variance model would draw two flat lines instead.

Estimation and Diagnostics

- No least squares route - σ_t^2 is never observed; parameters are chosen by maximum likelihood via the reconstructed recursion
- Report **robust** standard errors; ordinary ones are wrong under non-normal innovations
- Choose the innovation distribution deliberately - Student t (often $\hat{\nu} \in [4, 8]$) is the standard remedy for daily data

Diagnose the **standardised** residuals $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$: Ljung–Box on $\hat{z}_t$ checks the mean equation; on $\hat{z}_t^2$ checks the variance equation.

Asymmetry: When Good and Bad News Differ

Both ARCH and GARCH depend on ε_{t-1}^2 , discarding the sign - but equity falls raise subsequent volatility more than equal rises (the **leverage effect**).

$$\sigma_t^2 = \alpha_0 + (\alpha_1 + \delta \mathbb{I}[\varepsilon_{t-1} < 0]) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Positive shock has effect α_1 ; negative shock has effect $\alpha_1 + \delta$. Persistence becomes $\alpha_1 + \delta/2 + \beta_1 < 1$.

EGARCH and Very High Persistence

$$\ln \sigma_t^2 = \alpha_0 + \beta_1 \ln \sigma_{t-1}^2 + \alpha_1 (|z_{t-1}| - E|z_{t-1}|) + \delta z_{t-1}$$

Log form guarantees positivity without constraints; asymmetry enters through the **signed** shock δz_{t-1} .

When $\alpha_1 + \beta_1 = 1$: **IGARCH** - no long-run variance, forecasts never revert. Check for a structural break before accepting near-unit persistence at face value; splitting the sample and refitting often reveals the true story.

Spectral Analysis

Frequency, Period, and the Sampling Limit

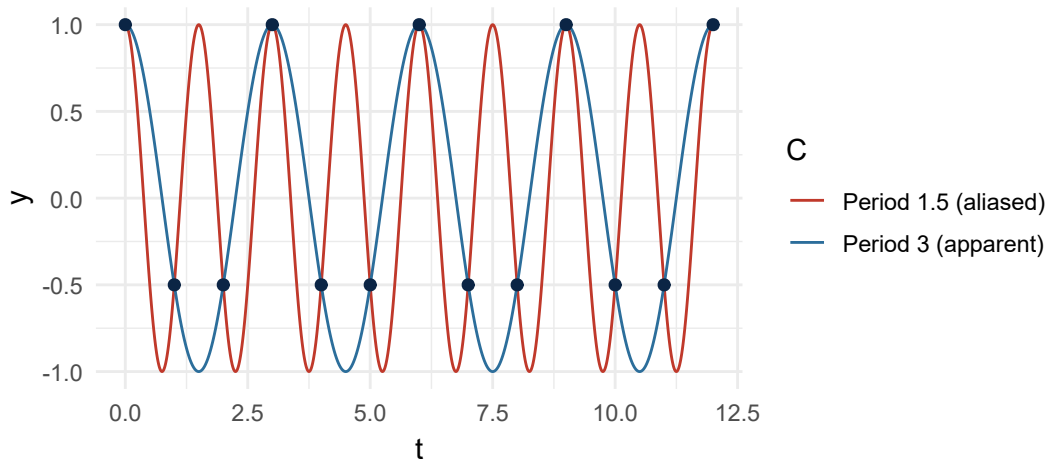
Nothing here competes with ARIMA - both domains describe the same process. A 12-month cycle buried in noise is diffuse in an ACF and a single spike in a periodogram; several superimposed cycles are unreadable in an ACF and immediately legible in a spectrum.

A sinusoid of period m has frequency $f = 1/m$ cycles/observation and angular frequency $\omega = 2\pi f$ radians/observation.

$$f_N = \frac{1}{2} \text{ cycle per observation.}$$

Oscillations faster than f_N are folded back (**aliasing**) - irreversible, since sampled values become numerically identical to a slower true frequency.

Aliasing



Sampled at the integers, periods 1.5 and 3 are indistinguishable - the information is absent, not merely obscured. Choose the sampling interval before collecting data.

A Series as a Sum of Waves

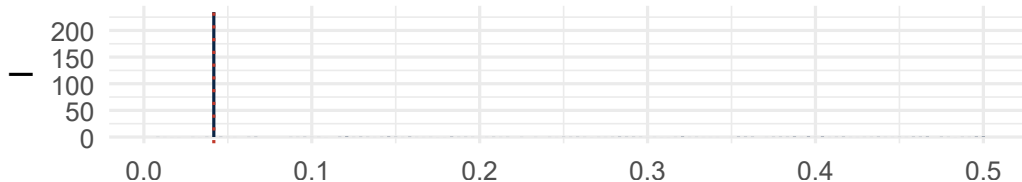
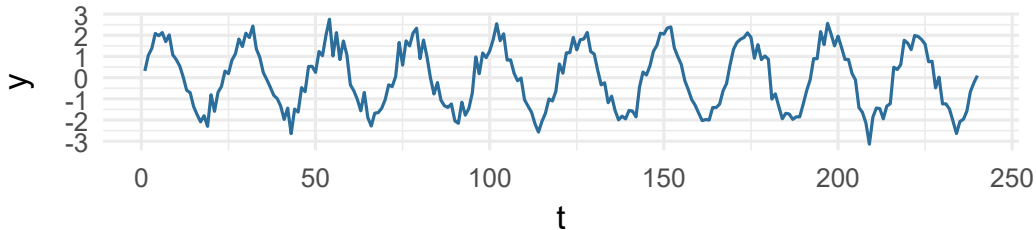
Any series can be written as a sum of sinusoids of different frequencies:

$$A \cos(2\pi ft + \varphi) = a \cos(2\pi ft) + b \sin(2\pi ft), \quad A = \sqrt{a^2 + b^2}.$$

Fitting at frequencies j/n makes the sine/cosine columns mutually orthogonal - the sum of squares splits cleanly across frequencies. The periodogram is that split, drawn as a picture: a change of coordinates, not a loss or gain of information.

Periodogram: First Example

$I(f)$ measures how much variance an oscillation at frequency f accounts for; total area under it is the sample variance.

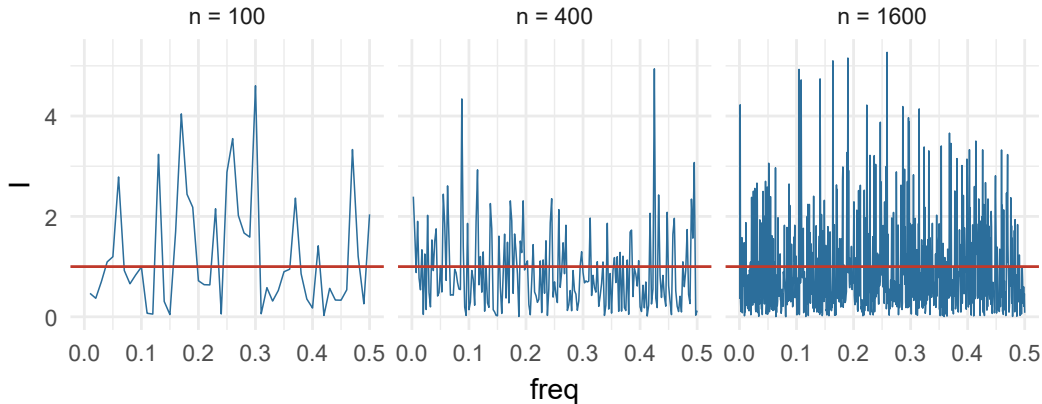


Reading a Periodogram: Five Steps

1. Locate the largest ordinate, convert $f \rightarrow 1/f$, attach units
2. Check for harmonics - a sharp seasonal spike (e.g. one December peak) needs many harmonics at $2/m, 3/m, \dots$, not separate cycles
3. Do not trust a single tall ordinate (next slide)
4. Smooth, and keep only what survives a wider span
5. Test the largest peak formally (Fisher's test) before reporting it

Raw Periodogram: Unbiased but Not Consistent

Each $I(f)$ estimates $S(f)$ correctly on average, but its SD is roughly its own mean - this does **not** shrink as n grows. More data gives more frequencies, each estimated just as badly.



Smoothing the Periodogram

`spec.pgram(x, spans = 9)` averages roughly nine neighbouring ordinates; `spans = c(9,7)` applies two passes.

- Too little smoothing: genuine features stay buried in noise
- Too much smoothing: peaks flatten, troughs fill in
- **A peak that vanishes when the span moves from nine to eleven was noise**
- Spectra are plotted on a log scale, with a single confidence bar of constant width

Fisher's Test, and Preparing a Series

$$g = \frac{\max_j I(f_j)}{\sum_j I(f_j)}$$

Referred to a null distribution accounting for having selected the **maximum** of many ordinates. Caveat: the null is white noise - on an already-autocorrelated series the test rejects readily; prewhiten (fit an ARMA, test the residuals) if the question is a cycle **beyond** ordinary serial correlation.

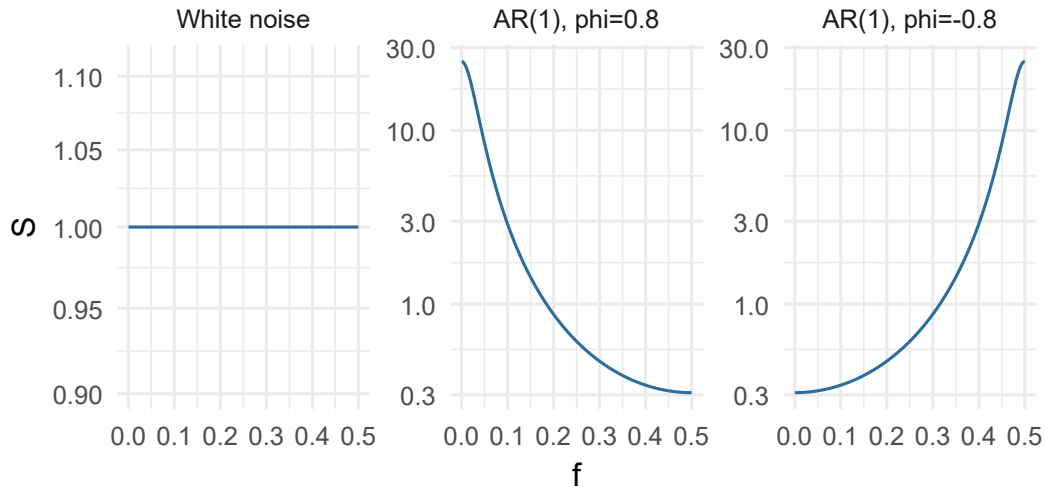
Three routine preparation steps (all `spec.pgram()` defaults): remove the mean/trend (else low-frequency power swamps everything); taper the ends (hard edges cause **leakage**; R defaults to 10%); padding is cosmetic - refines the frequency grid, adds no information.

Spectral Signatures of Familiar Processes

Process	Shape of $S(f)$
White noise	Flat everywhere
MA(1)	Monotone slope
AR(1), $\phi > 0$	Power at low frequencies (red noise)
AR(1), $\phi < 0$	Power at high frequencies (blue noise)
AR(2), complex roots	A peak at an interior frequency
Seasonal	Narrow spike at $1/m$ and harmonics

Stationarity is essential here too - a drifting covariance structure has no single $S(f)$ to estimate.

Model Spectra: Figure



An AR(2) with complex roots gives a **broad** interior peak - approximately regular cycles. unlike the narrow spike of true

What Differencing Does to a Spectrum

Filtering multiplies the spectrum, frequency by frequency, by a fixed curve:

- First difference $(1 - L)$: multiplier $4 \sin^2(\pi f)$ - zero at $f = 0$ (kills level/trend), **quadruples** power at $f = 1/2$
- Seasonal difference $(1 - L^{12})$: multiplier $4 \sin^2(12\pi f)$ - notches out the level and every seasonal harmonic exactly

Over-differencing amplifies the fastest, least informative variation - consistent with the earlier diagnostic that variance rises and $\rho(1) \approx -0.5$.

When to Reach for a Periodogram

Question	Time domain	Frequency domain
Is there dependence?	ACF/PACF	Departure from flat
How long is the cycle?	Awkward	Direct
Is the cycle regular?	Hard	Peak width answers it
Several cycles at once?	Very hard	Immediate
Forecast the next value	Directly	Only via a fitted model
Non-stationarity	Differencing	Neither domain applies

The frequency domain describes; it does not extrapolate - which is why ARIMA, not spectral analysis, carries most of a forecasting course.

Summary of the Course

- A time series is a stochastic process realised once; ordering carries the information
- Forecasting separates from goal-setting and planning; every forecast is a random variable
- Stationarity underpins the ACF/PACF, and the Box–Jenkins loop turns them into ARIMA/SARIMA models
- White noise is the standard every fitted model is judged against
- GARCH extends the standard to the conditional **variance**, where returns are uncorrelated yet highly predictable in magnitude
- Spectral analysis is the same information, read on a frequency axis instead of a time axis